

Design and Deployment of 3 -Tier Web Application on AWS Cloud

Introduction

This project demonstrates the implementation of a 3-tier architecture using Amazon Web Services (AWS). The architecture is divided into presentation, application, and database layers to ensure better security, scalability, and performance. Various AWS services such as VPC, EC2, Load Balancer, and RDS are used to build a highly available system. This project provides hands-on experience in designing real-world cloud infrastructure.

Objective

To design and deploy a scalable and secure 3-tier architecture using AWS cloud services.

To gain hands-on experience in implementing real-world cloud infrastructure and best practices .

Purpose

To gain practical knowledge of AWS services and implement real-world cloud architecture.

Architecture Overview

The project is designed using a 3-tier architecture model, which separates the application into three layers: Presentation Layer, Application Layer, and Database Layer.

This structure improves scalability, security, and maintainability

AWS Services Used

- **EC2 (Elastic Compute Cloud)** – Used to host frontend and backend servers
- **VPC (Virtual Private Cloud)** – Created network with public & private subnets
- **RDS (Relational Database Service)** – Used for database (MySQL/MariaDB)
- **S3 (Simple Storage Service)** – Used for storing static files (optional)
- **IAM (Identity and Access Management)** – Managed user access and permissions
- **Application Load Balancer** – Distributed traffic across servers
- **Auto Scaling** – Automatically scales instances based on load
- **Security Groups** – Controlled inbound and outbound traffic

AWS Services Used in 3-Tier Architecture

Networking

- VPC
- Public Subnet
- Private Subnet
- Internet Gateway
- NAT Gateway
- Route Tables

Presentation Layer (Frontend)

- Application Load Balancer (ALB)

Application Layer (Backend)

- EC2 Instances
- Auto Scaling Group
- Security Groups

Database Layer

- RDS (MySQL / MariaDB)

Security & Access

- IAM (Users, Roles, Policies)
- Security Groups

Implementation Steps

Create VPC

A Virtual Private Cloud (VPC) was created to establish a secure and isolated network environment in AWS. A custom CIDR block (10.0.0.0/16) was assigned to define the IP address range for the resources.

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - *optional*

Creates a tag with a key of 'Name' and a value that you specify.

anzee-tech

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

192.168.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

Figure: VPC Creation in AWS

VPC name

CIDR Block

Create Subnets

Subnets were created within the VPC to divide the network into smaller sections. Public subnets were used for internet-facing resources like the Load Balancer, while private subnets were used for application servers and the database to enhance security and control access.

VPC ID

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-012d60dbf208f2d7a (anzee-tech)

Associated VPC CIDRs

IPv4 CIDRs

192.168.0.0/16

Subnet Name, CIDR block, Availability Zone

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs

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Create Route Tables

Route tables were created to control the flow of network traffic within the VPC. A public route table was configured with a route to the Internet Gateway to allow external access. A private route table was configured to route traffic through a NAT Gateway, enabling secure outbound internet communication. Each subnet was associated with the appropriate route table.

Create route table [Info](#)
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs

Key	Value - optional	
Name	public route	Remove
Add new tag		

You can add 49 more tags.

Create Internet Gateway

An Internet Gateway was created and attached to the VPC to enable communication between resources in the VPC and the internet. A route was added in the public route table to allow internet access for resources in the public subnet.

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	int-gate	Remove

You can add 49 more tags.

Internet gateway name

Attach to VPC

Attach to VPC (igw-0cd9c1e54139cf412) Info

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

AWS Command Line Interface command

Create NAT Gateway

A NAT Gateway was created in the public subnet to allow instances in the private subnet to access the internet securely. It enables outbound internet communication while preventing direct inbound access from external sources.

NAT Gateway created in public subnet for private internet access.

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☒ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☐ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet level settings.

VPC
Select a VPC in which to create the regional NAT gateway.

Connectivity type
Select a connectivity type for the NAT gateway.

☒ **Public**

Public Subnet – Auto Assign Public IP

Auto-assign Public IP was enabled for the public subnet to automatically assign a public IP address to instances launched within it, allowing direct internet access.

VPC > Subnets > subnet-024fdcf268907c102 > Edit subnet settings

Edit subnet settings [Info](#)

Subnet
Subnet ID

Name

Auto-assign IP settings [Info](#)
Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ **Enable auto-assign public IPv4 address** [Info](#)

☐ **Enable auto-assign customer-owned IPv4 address** [Info](#)
Option disabled because no customer owned pools found.

Resource-based name (RBN) settings [Info](#)
Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ **Enable resource name DNS A record on launch** [Info](#)

☐ **Enable resource name DNS AAAA record on launch** [Info](#)

Public Route Table – Subnet Association

VPC > Route tables > rtb-024be3dbbb31a4ff1 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/3)

Filter subnet associations

< 1 > ⚙

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	public subnet	subnet-024fdcf268907c...	192.168.0.0/24	–	Main (rtb-037ad1ecc793504b7 / main)
☐	private subnet 01	subnet-01bb5c4bbc95b...	192.168.1.0/24	–	Main (rtb-037ad1ecc793504b7 / main)
☐	private subnet 02	subnet-0c3d2208e181f...	192.168.2.0/24	–	Main (rtb-037ad1ecc793504b7 / main)

Selected subnets

subnet-024fdcf268907c102 / public subnet ✕

Cancel

Save associations

Private Route Table – Subnet Association

VPC > Route tables > rtb-0122631223913305a > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/3)

Filter subnet associations

< 1 > ⚙

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	public subnet	subnet-024fdcf268907c...	192.168.0.0/24	–	rtb-024be3dbbb31a4ff1 / public route
☒	private subnet 01	subnet-01bb5c4bbc95b...	192.168.1.0/24	–	Main (rtb-037ad1ecc793504b7 / main)
☒	private subnet 02	subnet-0c3d2208e181f...	192.168.2.0/24	–	Main (rtb-037ad1ecc793504b7 / main)

Selected subnets

subnet-01bb5c4bbc95b30b4 / private subnet 01 ✕ subnet-0c3d2208e181f46b5 / private subnet 02 ✕

Cancel

Save associations

Add internet gateway in public route

VPC > Route tables > rtb-024be3dbbb31a4ff1 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes

Add NAT gateway in private route

VPC > Route tables > rtb-0122631223913305a > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes

Launch EC2 Instances

EC2 instances were launched in private subnets to host the application. These instances handle the business logic and process user requests coming from the load balancer.

Public server

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

public server

[Add additional tags](#)

Name

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

public-key01

[Create new key pair](#)

Create key pair

▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-012d60dbf208f2d7a (anzee-tech)
192.168.0.0/16



Subnet | [Info](#)

subnet-024fdcf268907c102
VPC: vpc-012d60dbf208f2d7a Owner: 250073196001
Availability Zone: us-east-2a (use2-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 192.168.0.0/24

public subnet

[Create new subnet](#)

subnet-024fdcf268907c102

Auto-assign public IP | [Info](#)

Enable

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

VPC ID, Public Subnet, Create security group

Private server

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

private server

[Add additional tags](#)

Name

▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-012d60dbf208f2d7a (anzee-tech)
192.168.0.0/16



Subnet | [Info](#)

subnet-01bb5c4bbc95b30b4 private subnet 01
VPC: vpc-012d60dbf208f2d7a Owner: 250073196001
Availability Zone: us-east-2b (use2-az2) Zone type: Availability Zone
IP addresses available: 251 CIDR: 192.168.1.0/24



[Create new subnet](#)

Auto-assign public IP | [Info](#)

Disable

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

VPC ID, Private subnet 01, existing security group

Public connection

```

    ~~~
    ~~-./
    _/m/'-/_/
Last login: Thu Apr  9 11:42:34 2026 from 3.16.146.5
[ec2-user@ip-192-168-0-45 ~]$ sudo passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-192-168-0-45 ~]$ su
Password:
[root@ip-192-168-0-45 ec2-user]# pwd
/home/ec2-user
[root@ip-192-168-0-45 ec2-user]# nano public-key01.pem
[root@ip-192-168-0-45 ec2-user]# ls
public-key01.pem
[root@ip-192-168-0-45 ec2-user]# chmod 400 "public-key01.pem"
[root@ip-192-168-0-45 ec2-user]# ssh -i "public-key01.pem" ec2-user@192.168.1.235
The authenticity of host '192.168.1.235 (192.168.1.235)' can't be established.
ED25519 key fingerprint is SHA256:3AfxdOvg+FWPvUwRzSYDEzXg9/kBod9wwntwTtCoKrK.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.235' (ED25519) to the list of known hosts.

```

Private connection

```
[root@cip 192.168.0.43 ~]# ssh -i /root/.ssh/id_rsa.pub ec2-user@192.168.1.235
The authenticity of host '192.168.1.235 (192.168.1.235)' can't be established.
ED25519 key fingerprint is SHA256:3AfxdOvg+FWPwVuRzSYDEzXg9/kBod9wwntwTtCoKrk.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.235' (ED25519) to the list of known hosts.
```

```
#  
~\##### Amazon Linux 2023  
~~\#####\  
~~\###|  
~~\#/  
~~V~'-'>  
~~~~  
~~.-.  
~~//  
~/m/'
```

[ec2-user@ip-192-168-1-235 ~]\$ sudo passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-192-168-1-235 ~]\$ su
Password:
[root@ip-192-168-1-235 ec2-user]#

Create new security group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Name, description, VPC ID

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
MySQL/Aurora	TCP	3306	Cust... sg-0a6db2668a41c9477	

Add rule

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info
MySQL/Aurora	TCP	3306	Cust... sg-0a6db2668a41c9477	

Inbound- MySQL/Aurora Source- security group (private server)

Outbound- MySQL/Aurora Source- security group (private server)

RDS

Create Subnet group

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

Data group01

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

remote

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

anzee-tech (vpc-012d60dbf208f2d7a)
3 Subnets, 3 Availability Zones

Name, description, VPC ID

Add subnets

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-2a

us-east-2b

us-east-2c

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

public subnet

Subnet ID: subnet-024fdcf268907c102 CIDR: 192.168.0.0/24

private subnet 01

Subnet ID: subnet-01bb5c4bbc95b30b4 CIDR: 192.168.1.0/24

private subnet 02

Subnet ID: subnet-0c3d2208e181f46b5 CIDR: 192.168.2.0/24

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Availability Zones (Three), Subnets (Three)

Create

Create databases

Engine options
Engine type [Info](#)

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☐ MySQL


☐ PostgreSQL


☒ MariaDB


☐ Oracle


☐ Microsoft SQL Server


☐ IBM Db2


Choose a database creation method

☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

MariaDB, Full configuration

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

anzee-tech (vpc-012d60dbf208f2d7a)
3 Subnets, 3 Availability Zones

Default VPC (vpc-0ec54cad4e25def17)
0 Subnets, 0 Availability Zones

☒ anzee-tech (vpc-012d60dbf208f2d7a)
3 Subnets, 3 Availability Zones

Create new VPC
Generates a new VPC automatically when you submit the form.

VPC ID

Existing VPC security groups

Choose one or more options

☐ secure-01

☒ rds secure

☐ default

New security group

Create

Endpoint RDS

Aurora and RDS

Dashboard

Databases

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

☐ Code snippets
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ CloudShell
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☒ Endpoints
Use when connecting through any IDE interface.

Master username
admin

Internet access gateway
Disabled

Port
3306

Endpoint type
Instance endpoint

▼ Additional configurations

Connectivity & security

Endpoint & port

Endpoint
database-1.cnqgimu0cwbp.us-east-2.rds.amazonaws.com

Port
3306

Networking

Availability Zone
us-east-2c

VPC
anzee-tech (vpc-012d60dbf208f2d7a)

Security

VPC security groups
rds-secure (sg-0e00ef805ffa45cc3)
Active

Publicly accessible
No

Connection

```
[ec2-user@ip-192-168-0-45 ~]$ sudo passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-192-168-0-45 ~]$ su
Password:
[root@ip-192-168-0-45 ec2-user]# dnf install mariadb105-server -y
Last metadata expiration check: 0:05:01 ago on Thu Apr  9 13:06:20 2026.
Package mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-192-168-0-45 ec2-user]# mariadb -hdatabase-1.cnqgimu0cwbp.us-east-2.rds.amazonaws.com -P 3306 -u admin -p --ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 87
Server version: 11.8.6-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Table created

```
MariaDB [school]> SHOW TABLES;
+-----+
| Tables_in_school |
+-----+
| students          |
+-----+
1 row in set (0.004 sec)

MariaDB [school]> INSERT INTO students (name, age) VALUES ('vikram', 21), ('surya', 22), ('vasu', 25);
Query OK, 3 rows affected (0.012 sec)
Records: 3 Duplicates: 0 Warnings: 0

MariaDB [school]> SELECT * FROM students;
+----+-----+-----+
| id | name  | age  |
+----+-----+-----+
| 1  | vikram | 21   |
| 2  | surya  | 22   |
| 3  | vasu   | 25   |
+----+-----+-----+
3 rows in set (0.002 sec)

MariaDB [school]>
```